

MILESTONE 2 REPORT

Predicting Student Performance Using OULAD
Hierarchical Classification: Fail | Good | Very Good | Excellent

Team ID: 60

ID	Name
2023170391	■■■■ ■■■■■■■■
2023170705	■■■■■ ■■■■ ■■■■ ■■■■■■
2023170283	■■■■ ■■■■■ ■■■■ ■■■■
2023170747	■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■
2023170749	■■■■■ ■■■■ ■■■■ ■■■■■■ ■■■■■■
2023170028	■■■■■ ■■■■ ■■ ■■■■ ■■■■■■

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1. Introduction & Project Evolution

Milestone 1 approached this dataset as a **regression problem**, predicting the numeric assessment score (0–100) using 30 engineered features, prior performance metrics, VLE click behaviour, and ensemble regressors (Random Forest, XGBoost, LightGBM, Stacking). The best result was $R^2=0.451$, $ACC@10=0.631$.

Milestone 2 reformulates the task as a **4-class classification problem**. Since the Milestone 2 dataset replaces the numeric score with a categorical *assessmentClass* column (Fail / Pass / Distinction / No Submission), regression is no longer appropriate. The target is engineered into four performance bands:

Class	Source	Meaning
Fail	final_result = Fail or Withdrawn	Did not successfully complete the module
Good	final_result = Pass (inconsistent)	Passed but with higher fail_rate or late_rate
Very Good	final_result = Pass (consistent)	Passed with fail_rate < 0.3 AND late_rate < 0.4
Excellent	final_result = Distinction	Highest performance band

A **hierarchical two-stage pipeline** was designed: Stage 1 predicts binary Pass/Fail, and its output is injected as an extra feature into Stage 2, which predicts the full 4-class label. This decomposition achieves **80.13% 4-class accuracy** with the champion HistGBM model.

2. Preprocessing Differences: Milestone 1 vs Milestone 2

The two milestones differ substantially in preprocessing strategy due to the fundamental change from a regression target (numeric score) to a classification target (assessmentClass).

2.1 Target Variable Construction

Aspect	Milestone 1 (Regression)	Milestone 2 (Classification)
Target variable	Numeric score (0–100)	4-class label: Fail / Good / Very Good / Excellent
Target source	studentAssessments.score (dropped rows with 'No Submission')	assessmentClass column; Withdrawn→Fail, Pass→Good/VeryGood, Excellent→Excellent
Missing target	Rows dropped (cannot impute regression target)	No Submission filled as 'No Submission' class; no rows dropped
Target engineering	None — raw score used directly	add_four_class_target() splits Pass into Good/VeryGood using frequency
Stage 1 target	N/A	Binary: Fail vs Pass (all passing classes collapsed to Pass)

2.2 Encoding Strategy

Feature group	Milestone 1	Milestone 2
Categorical (OHE)	code_module (removed), assessment_type, code_course, code_module, code_presentation, gender, region, disability — handled by OHE	code_module, code_course, code_presentation, gender, region, disability — handled by OHE
Categorical (Ord)	highest_education, imd_band, age_band (integer-encoded)	Same as Milestone 1 but OrdinalEncoder with unknown_value=-1 for unobserved categories
Numeric	MinMaxScaler on all 30 features after engineering	RobustScaler only for distance-based models (SVC, LR, SGD); t-SNE for visualization
Binary	disability and gender as 0/1	disability and gender kept as string categories, encoded by OHE
Rare category merge	No formal quals and Post-grad merged manually	Same merging inside reusable preprocess_df() function applied to all categorical features
Age band merge	Manual: 35-55 and 55+ → 35+	Same, inside preprocess_df()

2.3 Feature Selection & Noise Removal

Milestone 1 applied a structured 5-step feature selection pipeline ending with 30 features after correlation pruning and 5-fold stability selection. Milestone 2 takes a different approach: rather than formal selection, the Stage 2 ultimate model explicitly identifies and drops noisy features based on domain knowledge:

Dropped from Stage 2: studied_credits, module_presentation_length, num_of_prev_attempts, disability, age_band, gender, total_click — replaced by richer engineered signals (avg_hw_score, clicks_early, clicks_late, avg_days_late_TMA, avg_days_late_CMA).

3. Missing Value Handling Differences

Column / Issue	Milestone 1 Treatment	Milestone 2 Treatment
Score / assessmentClass	Rows with '?' dropped (target must be known)	assessmentClass NaN → 'No Submission' string class; no row
imd_band	Filled with column median (ordinal integer)	Filled per-region using the most frequent band from the training
date_registration	Filled with 0 for missing	Filled with date_unregistration minus training-set median offset.
total_click	Filled with 0	Filled with 0 (students with no VLE records had 0 interactions)
late_rate	Filled with 0	Filled with 1.0 (students with no submissions have 100% late ra
fail_rate	Filled with 0	Filled with 1.0 (students with no submissions have 100% fail rat
performance_level	N/A (not used in M1)	Filled with 'No Submission'; then dropped after 4-class target is
Missing input columns	Columns pre-built before modelling; no runtime	predict_from_raw_csvs() auto-joins missing columns from info/r
Train/test consistency	Scaler fit on train only (MinMaxScaler)	All imputation statistics (reg_date_median, imd_mode_map) fit

Key leakage prevention in Milestone 2: The add_four_class_target() function uses fail_rate and late_rate to assign the Very Good label, then both columns are dropped before any model sees them. Stage 2 training uses cross_val_predict on Stage 1 (not predict) so Stage 2 never sees inflated in-sample Stage 1 predictions.

4. Feature Engineering

4.1 Base Features (both milestones)

Feature	Source	Description
total_click	studentVle	Total VLE interactions per student per module-presentation
late_rate	studentAssessments	Proportion of submitted assessments submitted after deadline
fail_rate	studentAssessments	Proportion of assessed submissions classified as Fail
performance_level	studentAssessments	Mode assessmentClass per student per module (most common outcome)
date_registration	studentRegistration	Days relative to module start when student registered
module_presentation_length	courses	Duration of the module in days
num_of_prev_attempts	studentInfo	Number of times student previously attempted this module
studied_credits	studentInfo	Total credits student is studying in this presentation
imd_band	studentInfo	Index of Multiple Deprivation band (socioeconomic indicator)

4.2 Advanced Features Engineered in Stage 2 (Milestone 2 only)

Feature	Engineering Logic	Why it helps
avg_hw_score	Mean numeric score across all non-exam assignments (5=Fail, 0=Pass, 1=Very Good, 2, Excellent=3)	Capable of predicting final outcome
clicks_early	Sum of VLE clicks before day 50 of module	Early engagement strongly predicts final outcome
clicks_late	Sum of VLE clicks from day 50 onward	Sustained engagement vs cramming pattern distinction
avg_days_late_TMA	Mean days late across Tutor-Marked Assignments	Submission timing per assessment type
avg_days_late_CMA	Mean days late across Computer-Marked Assignments	Different behaviour for automated vs tutor-marked work
coarse_pred	Stage 1 binary prediction (0=Fail, 1=Pass)	Gives Stage 2 a strong prior about the student's broad outcome

5. Hierarchical Classification Pipeline

The core architectural decision in Milestone 2 is a two-stage hierarchical pipeline. Rather than training a single model to distinguish all four classes simultaneously (which was observed to produce ~56% accuracy for Decision Trees), the problem is decomposed into two easier subproblems:

Stage 1 — Binary Classifier (Pass vs Fail)

All four raw labels are collapsed to two: Fail (includes Withdrawn) and Pass (includes Good, Very Good, Excellent). This is an inherently easier problem with clear feature separation and achieves **83.82% accuracy**. Three models are evaluated: ExtraTrees, HistGBM (tuned), and Random Forest. HistGBM is selected as the Stage 1 champion.

Model	CV Accuracy	Notes
ExtraTrees	~0.82	class_weight='balanced', n_estimators=200
HistGBM	~0.838	CHAMPION — tuned via RandomizedSearchCV, n_iter=40
RandomForest	~0.82	class_weight='balanced', n_estimators=200

Stage 2 — 4-class Classifier with Coarse Feature Injection

The Stage 1 binary prediction (coarse_pred = 0 or 1) is appended as an additional numeric feature to the original feature matrix. Stage 2 then trains on this augmented matrix to predict the fine-grained 4-class label. **cross_val_predict** is used on the training set (not predict) to generate out-of-fold Stage 1 predictions — this is critical to prevent leakage where an overfitted Stage 1 provides artificially clean predictions to Stage 2.

Stage 2 Ultimate — Advanced Feature Engineering + Noise Dropping

An enhanced Stage 2 pipeline adds five new features (avg_hw_score, clicks_early, clicks_late, avg_days_late_TMA, avg_days_late_CMA) and drops seven noisy columns identified as low-signal for 4-class discrimination. Three models are tuned with RandomizedSearchCV (n_iter=5, cv=3):

Model	Best CV Acc	Best Params
ExtraTrees	0.7662	n_estimators=100, min_samples_split=5, max_depth=None
RandomForest	0.7870	n_estimators=100, min_samples_leaf=2, max_depth=15
HistGBM	0.8042	max_iter=100, max_depth=5, lr=0.1, l2=0.0 ■ CHAMPION

6. Model Results & Comparison

6.1 Stage 1 — Binary Classification Results

Metric	Value
Test Accuracy	83.82%
Precision (Fail)	89%
Recall (Fail)	80%
F1 (Fail)	84%
Precision (Pass)	80%
Recall (Pass)	88%
F1 (Pass)	84%
Macro F1	84%
Weighted F1	84%

6.2 Stage 2 — 4-class Champion Results (HistGBM)

Class	Precision	Recall	F1-Score	Support
Excellent	0.60	0.44	0.51	613
Fail	0.92	0.89	0.90	3447
Good	0.67	0.69	0.68	1122
Very Good	0.71	0.84	0.77	1337
Accuracy			0.8013	6519
Macro avg	0.72	0.71	0.71	6519
Weighted avg	0.80	0.80	0.80	6519

6.3 Improvement Over Baseline

The hierarchical pipeline with the ultimate Stage 2 represents a substantial improvement at each stage:

Pipeline Stage	Accuracy	Improvement vs Flat 4-class
Flat ExtraTrees (4-class)	~67%	baseline
Flat HistGBM (4-class)	~71%	+4%
Hierarchical (basic)	~70%	+3%
Hierarchical + Advanced Features	80.13%	+13% over flat baseline ■

6.4 Confusion Matrix Analysis

From the Stage 2 confusion matrix with 6,519 test samples:

- Fail class: 3,057 correctly classified out of 3,447 (88.7% recall). Only 140 Fail students misclassified as Very Good and 234 as Good — suggesting a subset of failing students still maintained moderate VLE engagement.

- Very Good class: 1,121 / 1,337 correct (83.8% recall). Best-recalled passing class, confirming that the feature set clearly captures the consistent, high-engagement student profile.
- Good class: 776 / 1,122 correct (69.1% recall). Most confusion is with Very Good (73 students), indicating the Good/Very Good boundary is the hardest to separate.
- Excellent class: Only 270 / 613 correct (44.0% recall). This is the hardest class — Distinction students are a small minority and their behaviour partially overlaps with Very Good students.

7. Full Model Report

7.1 Dataset Summary

Attribute	Value
Source dataset	OULAD — Open University Learning Analytics Dataset
Files used	assessments.csv, studentAssessments.csv (with assessmentClass), studentInfo.csv, studentRegistrations.csv
Total students	~32,593 unique student-module-presentation records
Training samples	26,074
Test samples	6,519
Split ratio	80/20 stratified by final_result
Target classes	Fail (52.8%), Very Good (20.5%), Good (17.2%), Excellent (9.4%)
Base features	5 OHE categorical + 3 ordinal + 5 numeric = 13 raw; expands to 36 after encoding
Stage 2 final features	11 after noise drop + 5 advanced engineered + coarse_pred = 17 input columns

7.2 Top Predictors (Permutation Importance — Stage 2)

Rank	Feature	Importance	Notes
1	total_click	0.170	VLE engagement is the single strongest signal
2	code_module	0.103	Module identity — different modules have very different pass rates
3	coarse_pred	0.054	Stage 1 binary prediction — powerful hierarchical feature
4	code_presentation	0.027	Semester affects difficulty and pass rates
5	region	0.008	Geographic region has minor but measurable effect
6	date_registration	0.003	Early registration correlates weakly with engagement
7–14	Other features	<0.003	num_of_prev_attempts, studied_credits, education, gender, etc.

7.3 Key Technical Design Decisions

Decision	Rationale
Hierarchical 2-stage pipeline	Flat 4-class models achieved only ~67% accuracy. Decomposing into binary (Stage 1) + fine-grained (Stage 2) improved performance.
Using cross_val_predict for Stage 1 train prediction	Using predict() directly would cause leakage — Stage 1 would have memorised training data, giving artificially high accuracy.
HistGradientBoostingClassifier as champion	10-50x faster than GradientBoostingClassifier (histogram binning like LightGBM). Native missing value support.
RobustScaler instead of MinMaxScaler	Total_click is heavily right-skewed (outliers up to 25,000+). RobustScaler uses median/IQR instead of mean/std.
Dropping noisy features in Stage 2	studied_credits, module_presentation_length, num_of_prev_attempts, disability, age_band, gender were dropped as they added noise without predictive power.
preprocess_df() reusable function	All imputation (imd_band regional mode, date_registration median) is encapsulated in one function for consistency.
add_four_class_target() with drop-after	fail_rate and late_rate are used only to assign the Very Good label, then immediately dropped. This prevents leakage of Stage 2 targets into Stage 1.

7.4 Saved Artifacts

File	Size	Contents
stage1_binary_pipeline.pkl	~547 KB	Fitted HistGBM pipeline for Pass/Fail binary prediction
stage2_4class_pipeline.pkl	~5.3 MB	Fitted HistGBM pipeline for 4-class prediction
stage2_ultimate_champion.pkl	variable	Champion model with advanced features
stage2_extratrees.pkl	~6 KB	ExtraTrees Stage 2 (unfitted shell — refit before use)
stage2_randomforest.pkl	~6 KB	RandomForest Stage 2
stage2_sgd.pkl	~2 KB	SGD Stage 2
preprocessing_bundle.pkl	small	reg_date_median, imd_mode_map, column lists, coarse_map

7.5 Inference Pipeline

The `predict_from_raw_csvs()` function accepts the 6 raw CSV files directly and returns a prediction DataFrame with no preprocessing required from the caller:

Step	Action
1	Load all CSV files (accepts file paths or already-loaded DataFrames)
2	Build <code>total_click</code> from <code>studentVle</code> + <code>vle</code> (if provided; else 0)
3	Build <code>fail_rate</code> , <code>late_rate</code> , <code>performance_level</code> from assessment submission records
4	Rebuild <code>regCoursesInfo</code> from <code>reg</code> + <code>courses</code> + <code>info</code> (<code>studentInfo</code>) tables
5	Merge everything into one feature matrix
6	<code>preprocess_df()</code> with training-set statistics from <code>preprocessing_bundle.pkl</code>
7	<code>add_four_class_target()</code> → drop leaky columns (<code>fail_rate</code> , <code>late_rate</code>)
8	Enforce column types (str for categoricals, float for numerics)
9	Stage 1: <code>s1_model.predict(X)</code> → <code>coarse_pred</code>
10	Stage 2: <code>s2_model.predict(X + coarse_pred)</code> → final 4-class prediction + probabilities
11	Return DataFrame: <code>id_student</code> , <code>code_module</code> , <code>code_presentation</code> , <code>stage1_coarse</code> , <code>predicted_class</code> , <code>proba_*</code>

8. Discussion & Conclusion

8.1 Comparison with Milestone 1

Dimension	Milestone 1	Milestone 2
Task type	Regression	4-class Classification
Target variable	Numeric score 0–100	Fail / Good / Very Good / Excellent
Best metric	$R^2=0.451$, $ACC@10=63.1\%$	Accuracy=80.13%, Macro F1=71%
Best model	ACC-Optimized Blend (RF+XGB+LGB+Stacking)	Hierarchical HistGBM (Stage 1 + Stage 2)
Feature count	30 selected features	17 (Stage 2 ultimate after noise drop)
Strongest predictor	prior_mean_score ($r=0.42$)	total_click (perm. importance=0.170)
Leakage prevention	Scaler fit on train only	Scaler + imputation stats + cross_val_predict for Stage 1
Missing value strategy	0-fill or median-fill	Domain-informed: late_rate→1.0, imd_band→regional mode
Deployment artifact	No saved artifacts	preprocessing_bundle.pkl + 6 saved model files

8.2 Key Findings

- The hierarchical approach is definitively better than flat 4-class classification: +13% accuracy over a flat ExtraTrees baseline by injecting Stage 1's binary prediction as a feature.
- The Excellent class remains the hardest to predict (44% recall) because Distinction students form a small minority (9.4%) and behaviourally overlap significantly with Very Good students.
- VLE engagement (total_click) is the most predictive feature in Milestone 2, confirming Milestone 1's finding that behavioural features outperform demographics.
- The module identity (code_module) is the second strongest predictor — different modules have fundamentally different pass rates, suggesting per-module models could further improve accuracy.
- The coarse_pred feature from Stage 1 ranks 3rd in permutation importance, validating the hierarchical decomposition: giving Stage 2 a binary prior dramatically sharpens its 4-class predictions.
- Advanced time-aware features (clicks_early vs clicks_late) and per-assessment-type late submission averages provided the +9% lift from Stage 2 basic (0.70) to Stage 2 ultimate (0.80).

8.3 Limitations

- Excellent class recall is only 44% — the model struggles to identify the top-performing students reliably. A dedicated Excellent vs Very Good classifier or SMOTE oversampling could address this.
- The Very Good class is defined by a threshold on fail_rate/late_rate, which are dropped before training. The model infers Very Good indirectly from VLE behaviour and module identity.
- No temporal features (time-series of daily clicks) are used — a recurrent or sequence model capturing the trajectory of engagement could substantially improve predictions.
- The inference pipeline requires access to the original studentVle.csv to build total_click — without it, click features default to 0 and accuracy would be lower.

8.4 Next Steps

- XGBoost or LightGBM with native categorical support could further improve Stage 2 accuracy.
- A dedicated Stage 3 model separating Excellent from Very Good students (binary sub-classifier) could improve Excellent recall from 44% to 70%+.

- Per-module separate models — since code_module is the 2nd strongest predictor, training one model per module could capture module-specific patterns.
- Time-series features: rolling click counts per week, early-semester engagement slope, assignment completion rate over time.
- SMOTE or class-weight tuning specifically for the Excellent minority class.

Final Champion: Hierarchical HistGBM — Stage 1: 83.82% binary accuracy | Stage 2: 80.13% 4-class accuracy

Team 60 — Milestone 2 Report